Day 085 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Compare Rate Models and Units

Standards: 6.RP.A.2, 6.RP.A.3b

Learning target: I can compare Rate Models and Units, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A unit rate compares an amount with one unit of another quantity. Divide the first quantity by the second. Write the units, such as dollars per notebook, so you know what the quotient means. Divide total cost by the number of items or units. Compare costs for exactly one of the same unit. A lower package price does not always give a lower unit price. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2, 21

Day 085 | Unit 06 | Lesson

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01 / Learn

A unit rate compares an amount with one unit of another quantity. Divide the first quantity by the second. Write the units, such as dollars per notebook, so you know what the quotient means. Divide total cost by the number of items or units. Compare costs for exactly one of the same unit. A lower package price does not always give a lower unit price. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating.

Worked example 1

7 identical notebooks cost $49. Find the cost per notebook.

1. A unit rate compares an amount with one unit of another quantity.

2. Divide the first quantity by the second.

3. Unit rate = 49 ÷ 7 = 7 dollars per notebook.

4. State the result in the form requested: $7 per notebook.

Answer: $7 per notebook

Worked example 2

A bicycle travels 140 km in 4 hours at constant speed. Select all true statements.
A. In 1 hour it travels 36 km.
B. The speed is 35 km per hour.
C. The speed is 144 km per hour.
D. In 3 hours it travels 105 km.

1. At a constant speed, distance = speed × time.

2. Divide distance by time: 140 ÷ 4 = 35 km/h.

3. Multiply this unit rate by 3 for the later distance.

Answer: B, D - The speed is 35 km per hour.; In 3 hours it travels 105 km.

Day 085 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

4 identical notebooks cost $16. Find the cost per notebook.

02 / Constructed response

Pack A has 3 markers for $18; Pack B has 6 markers for $39. Find each unit price and the better buy.

03 / Constructed response

A cyclist travels 80 km in 2 hours at constant speed. Find the speed.

04 / Constructed response

4 identical notebooks cost $28. Find the cost per notebook.

05 / Constructed response

Pack A has 4 markers for $16; Pack B has 9 markers for $40.5. Find each unit price and the better buy.

06 / Constructed response

A bus travels at 10 km/h. How long does 30 km take?

07 / Constructed response

2 identical notebooks cost $6. Find the cost per notebook and the cost of 5 notebooks at this rate.

08 / Two-part numeric entry and reasoning

Store A sells 6 notebooks for $12. Store B sells 7 notebooks for $21. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

09 / Multiple select

A bicycle travels 70 km in 2 hours at constant speed. Select all true statements.
A. The speed is 35 km per hour.
B. In 4 hours it travels 140 km.
C. In 1 hour it travels 36 km.
D. The speed is 72 km per hour.

10 / Constructed response

4 identical notebooks cost $14. Find the cost per notebook and the cost of 7 notebooks at this rate.

Day 085 | Unit 06 | Answer key

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1. $4 per notebook

Unit rate = 16 ÷ 4 = 4 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

2. A $6/marker; B $6.5/marker; A is cheaper

Divide each price by its marker count; compare 6 with 6.5.

6.RP.A.3b | Constructed response

3. 40 km/h

Speed = 80 ÷ 2 = 40; distance = speed × time.

6.RP.A.3b | Constructed response

4. $7 per notebook

Unit rate = 28 ÷ 4 = 7 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

5. A $4/marker; B $4.5/marker; A is cheaper

Divide each price by its marker count; compare 4 with 4.5.

6.RP.A.3b | Constructed response

6. 3 hours

Time = distance ÷ speed = 30 ÷ 10 = 3.

6.RP.A.3b | Constructed response

7. $3 per notebook; $15

Unit rate = 6 ÷ 2 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

8. A: $2 each; B: $3 each; A is the better buy.

Divide each cost by its own notebook count: 12/6 = 2; 21/7 = 3. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

9. A, B - The speed is 35 km per hour.; In 4 hours it travels 140 km.

Divide distance by time: 70 ÷ 2 = 35 km/h. Multiply this unit rate by 4 for the later distance.

6.RP.A.3b | Multiple select

10. $3.5 per notebook; $24.5

Unit rate = 14 ÷ 4 = 3.5 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response